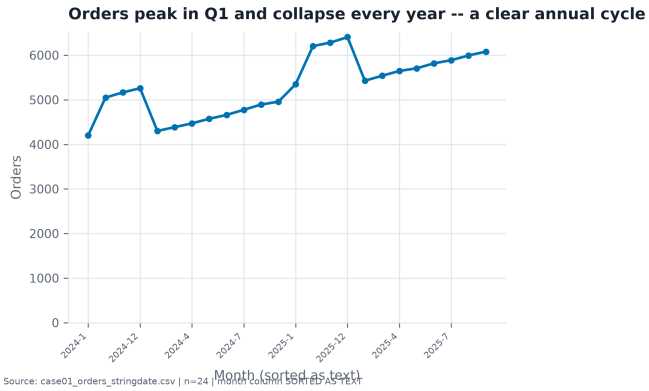
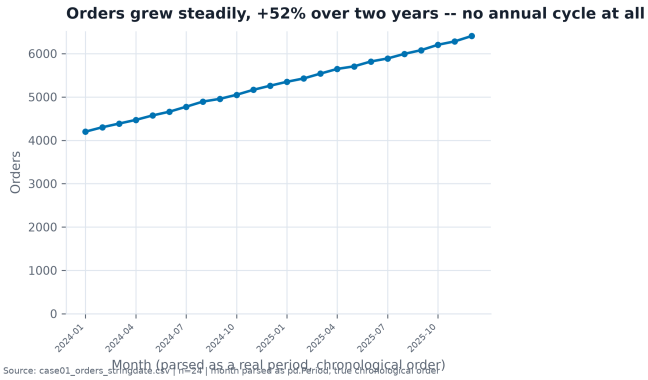
Part A – Nine Diagnoses

Week 4 Data Visualisation Assignment | Roll No: BSAI-010

1. Case 01 Orders String-Date





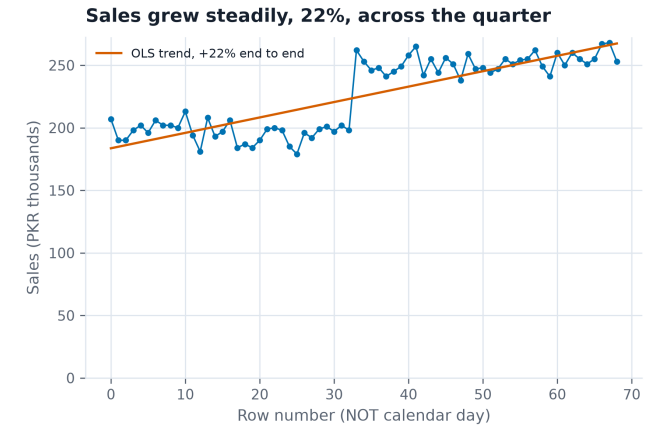
Invited conclusion: Orders peak in Q1 and collapse every year – a clear annual cycle.

Decoy objection ruled out: Could the sample be too small? No – n=24, a full two years of monthly data; the issue is order, not size.

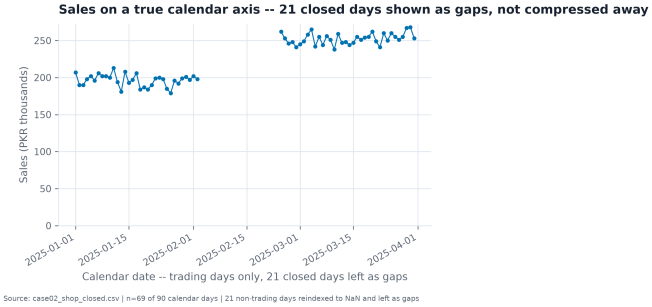
Mechanism: The month column is text (‘2024-1’, ‘2024-10’, ...) with no zero-padding. Sorting it as a STRING is not chronological: ‘2024-10’ < ‘2024-2’ lexicographically, so October, November and December sort before February. The series genuinely climbs monotonically; the two ‘drops’ (958 and 978) are artefacts of the resort, not business events.

Redesign: Parse the month as a pd.Period and sort chronologically – orders rise steadily +52%, no cycle.

2. Case 02 Shop Closed



Source: case02_shop_closed.csv | n=69 trading-day rows | x = row number



Source: case02_shop_closed.csv | n=69 of 90 calendar days | 21 non-trading days reindexed to NaN and left as gaps

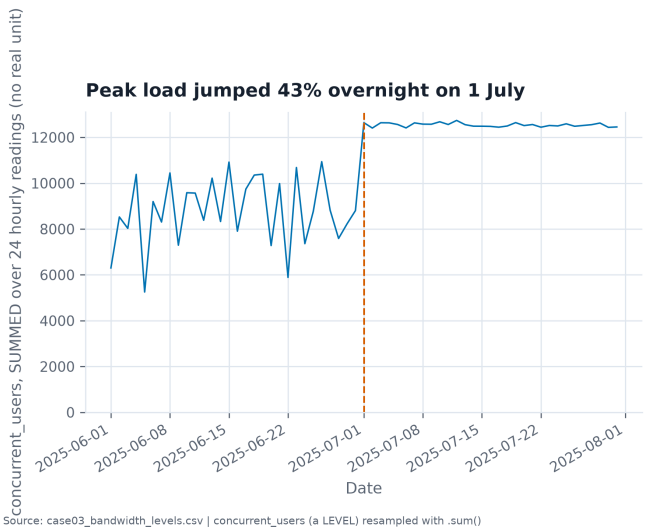
Invited conclusion: Sales grew steadily, 22%, across the quarter.

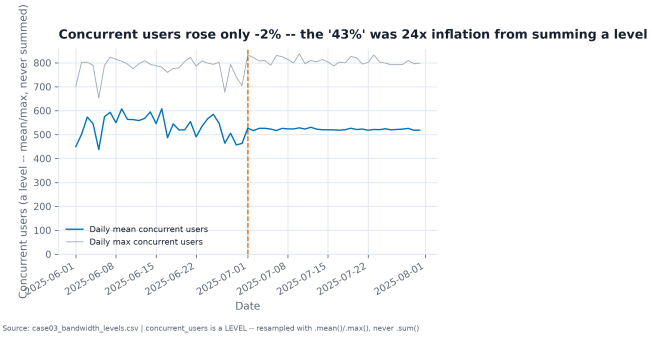
Decoy objection ruled out: Could the OLS line be overfit? No – the trend is real; the issue is the x-axis, not the regression.

Mechanism: The file only records trading days (weekends excluded). Plotting against row number treats 69 consecutive rows as 69 consecutive days, silently compressing the real 90-day calendar span and erasing the weekend gaps – a reader cannot see that Friday-to-Monday is one business day’s gap on this axis but three calendar days.

Redesign: Reindex onto a full calendar and plot by true date, leaving closed days as gaps in the line.

3. Case 03 Bandwidth Levels





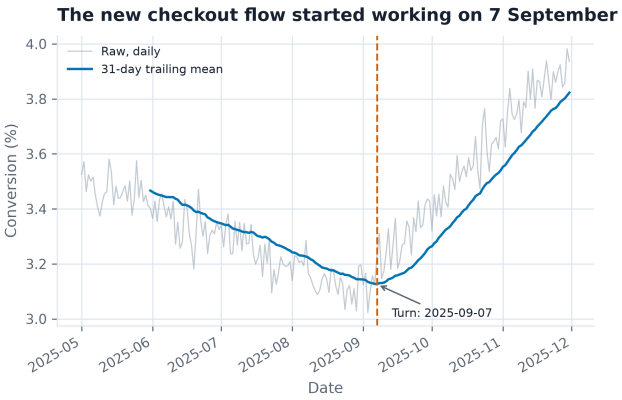
Invited conclusion: Peak load jumped 43% overnight on 1 July.

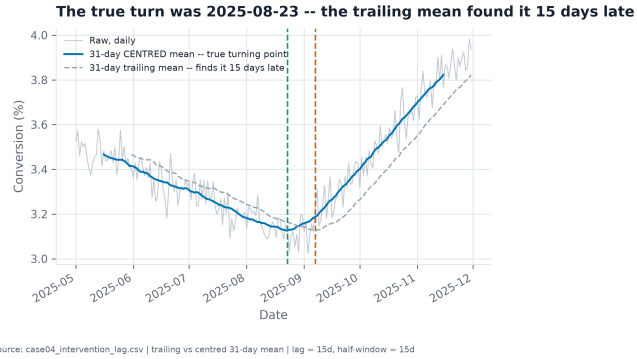
Decoy objection ruled out: Could July simply have more users? Real growth would show in the MEAN level too – and it barely moves.

Mechanism: concurrent_users is a LEVEL (a snapshot of who is connected at that instant), not a flow. Resampling a level with .sum() adds up 24 unrelated snapshots per day, a number with no real-world meaning. The ‘43% jump’ is sensitive to how many hours of already-elevated readings happen to fall in July vs June, not a step change in capacity.

Redesign: Resample the level with .mean() or .max(), never .sum() – the true change is a few percent.

4. Case 04 Intervention Lag





Source: case04_intervention_lag.csv | 31-day TRAILING rolling mean

Source: case04_intervention_lag.csv | trailing vs centred 31-day mean | lag = 15d, half-window = 15d

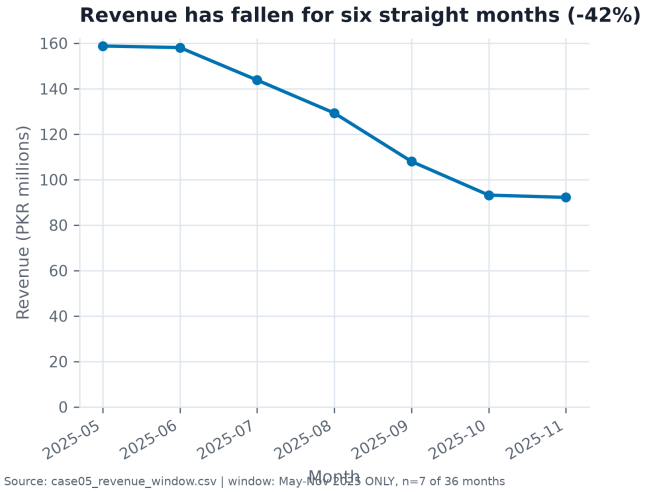
Invited conclusion: The new checkout flow started working on 7 September.

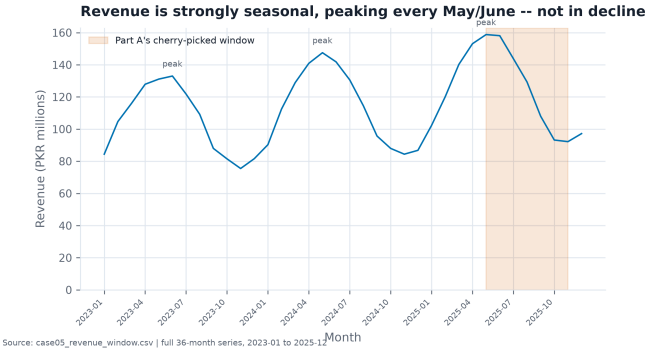
Decoy objection ruled out: Could the 31-day window be too long? No – any trailing window has the same lag problem, just a different number of days late.

Mechanism: A trailing rolling mean at day t only knows about days t-30..t, so it cannot register a turn until the window has filled. lag_at_turn() shows the trailing mean finds the minimum roughly half a window (about 15 days) after the centred mean does – the real turning point is in late August, not 7 September.

Redesign: A centred rolling mean locates the true turn; report both and state which a live dashboard can show.

5. Case 05 Revenue Window





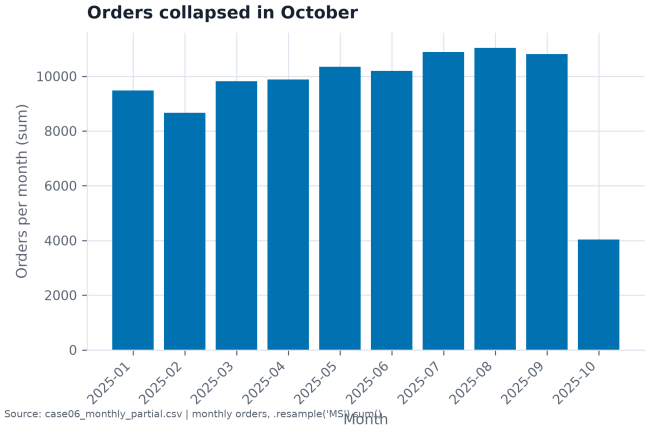
Invited conclusion: Revenue has fallen for six straight months.

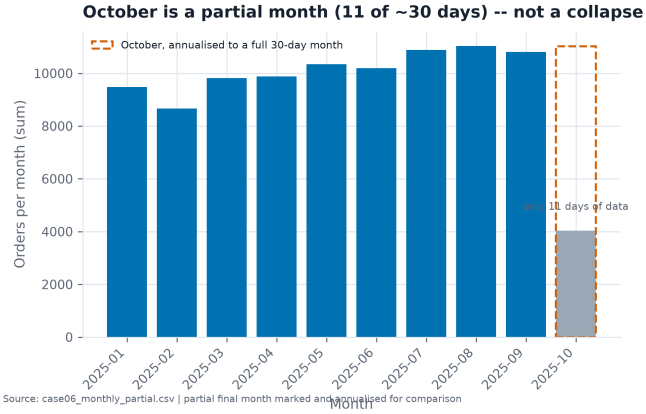
Decoy objection ruled out: Is n=7 too small to trust? n is fine for describing those 7 months – the problem is which 7 months were chosen, not how many.

Mechanism: The full 36-month series is strongly seasonal, peaking every May/June and troughing every November, repeated three times. May-to-November is exactly the down-leg of one cycle. The same ‘6 straight months of decline’ claim could be made every single year on this series.

Redesign: Plot the full 36 months with the cherry-picked window and the three annual peaks marked.

6. Case 06 Monthly Partial





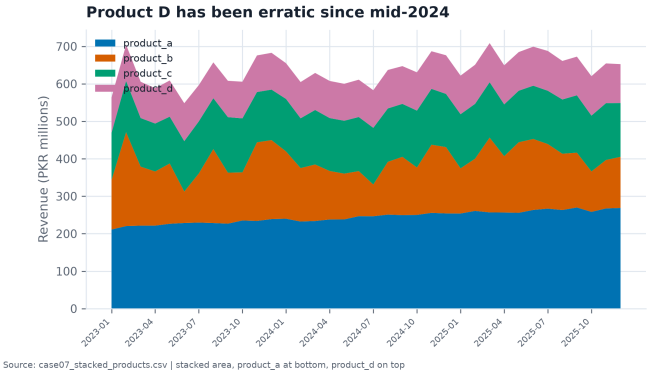
Invited conclusion: Orders collapsed in October.

Decoy objection ruled out: Could October genuinely be a weak month? The daily run-rate (orders per day) is actually in line with September once the shorter month is accounted for.

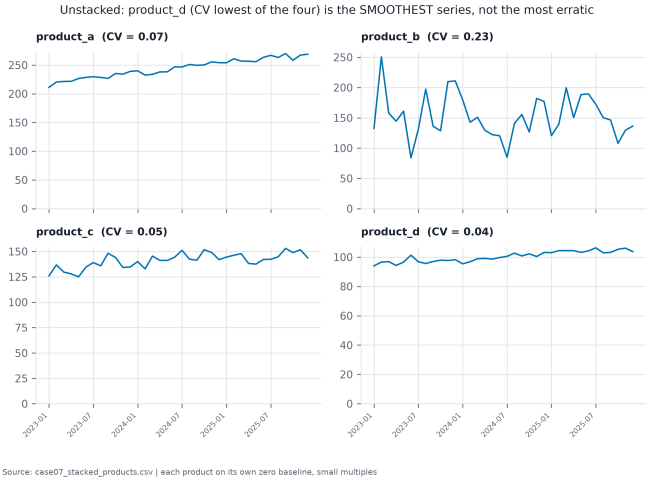
Mechanism: The extract was taken part-way through October, so the October bar sums far fewer days than every complete month before it. A partial period summed and drawn at the same visual weight as complete periods always looks like a collapse.

Redesign: Mark the partial month distinctly and show its run-rate annualised to a full month for comparison.

7. Case 07 Stacked Products



Source: case07_stacked_products.csv | stacked area, product_a at bottom, product_d on top



Source: case07_stacked_products.csv | each product on its own zero baseline, small multiples

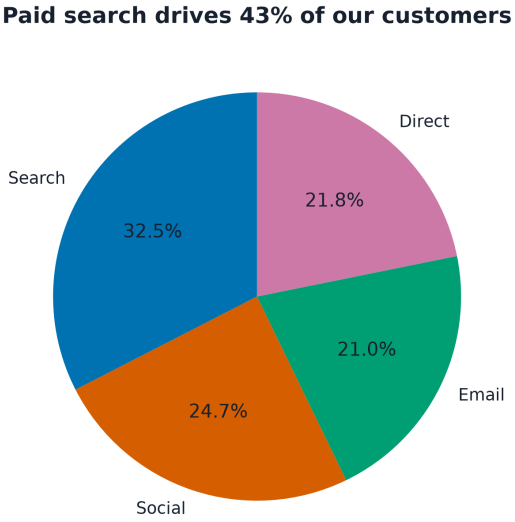
Invited conclusion: Product D has been erratic since mid-2024.

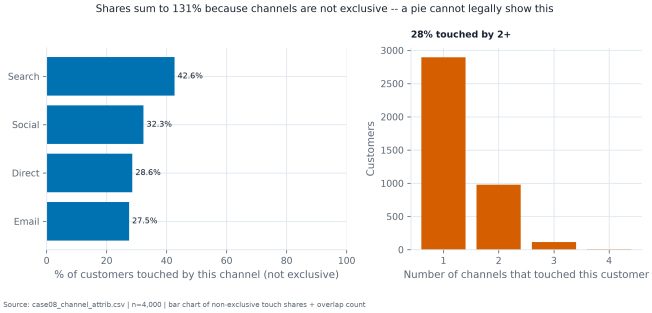
Decoy objection ruled out: Could product_d's own numbers be volatile? Its coefficient of variation is the LOWEST of the four products – it is the smoothest, not the most erratic.

Mechanism: Volatility leaking upward through a stacked area's moving baseline. product_d is drawn on top of product_a+b+c, so its visible band wobbles whenever the series beneath it (chiefly the volatile product_b) wobbles, even though product_d's own value barely moves.

Redesign: Unstack into small multiples, each product on its own zero baseline – product_d is visibly the calmest series.

8. Case 08 Channel Attribution





Source: case08_channel_attrib.csv | n=4,000 | PIE OF NON-EXCLUSIVE SHARES summing to 131.0%

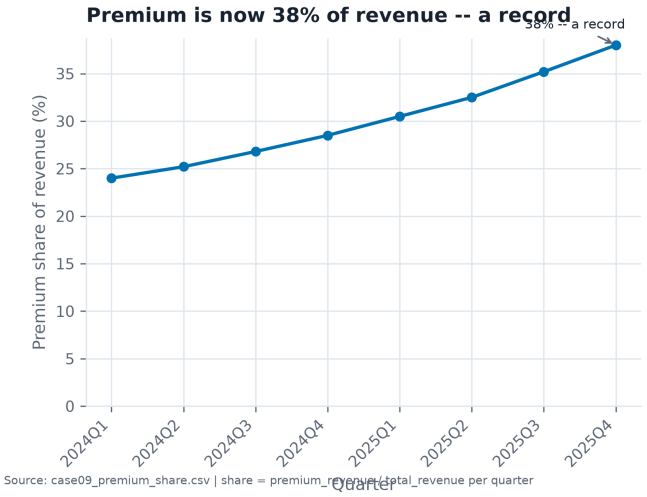
Invited conclusion: Paid search drives 43% of our customers.

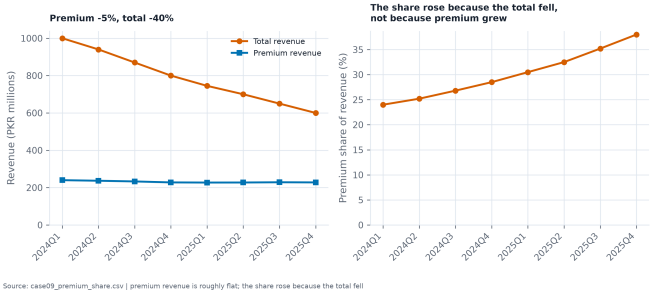
Decoy objection ruled out: Could the 43% simply be rounded from a clean number? No – the four shares sum to well over 100%, which is impossible for a true partition.

Mechanism: exclusivity_check() shows the four touch_* shares sum to over 130% and a large share of customers are touched by more than one channel before purchase – the categories are not mutually exclusive, so a pie (or any part-to-whole chart) of them is not legal.

Redesign: A sorted bar chart of (non-exclusive) touch shares, plus a histogram of how many channels touched each customer.

9. Case 09 Premium Share





Invited conclusion: Premium is now 38% of revenue – a record.

Decoy objection ruled out: Did premium revenue actually grow? It is essentially flat quarter to quarter (+240 to +228, even dipping) – the growth is not there.

Mechanism: A rising share of a falling total. Premium revenue barely changes across the 8 quarters while total revenue nearly halves; the share rises mechanically as the denominator shrinks, not because the premium line is winning more business.

Redesign: Two series (premium and total revenue) plotted directly, beside the share – showing the share rose for the wrong reason.